

Two-Channel WIRELESS AUDIO AMPLIFIER Using Bluetooth And TA8210AH

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Use of audio amplifiers is increasing day by day. This project deals with the transmission of the audio signal wirelessly and amplification of the received signal using VTF-108 Bluetooth module and audio amplifier IC TA8210AH.

Bluetooth module can connect to devices such as Hi-Fi equipment,

mobile phones and MP3 players. Audio sources such as mobilephones work as the transmitter unit, and Bluetooth module, audio amplifier and speakers form the receiver or wireless speaker unit. The receiver unit can be positioned anywhere (within Bluetooth range), so it can be moved around without using any cables. Amplifier in the receiver unit is used to boost the audio signal.

This design can be used in four-wheel vehicles, homes, seminar halls and so on. The author's prototype is shown in Fig. 1.

Circuit and working

Circuit diagram of the 2-channel wireless audio amplifier is shown in Fig. 2. It is built around 2-channel audio amplifier TA8210AH (IC2), VTF-108 Bluetooth module, two

loudspeakers (LS1 and LS2), resistors (R1 through R7), capacitors (C1 through C11) and some other components. VTF-108 module requires 5V DC power supply, whereas audio amplifier IC2 uses 12V DC power supply.

The whole design requires the connections as described below.



Fig. 1: Author's prototype

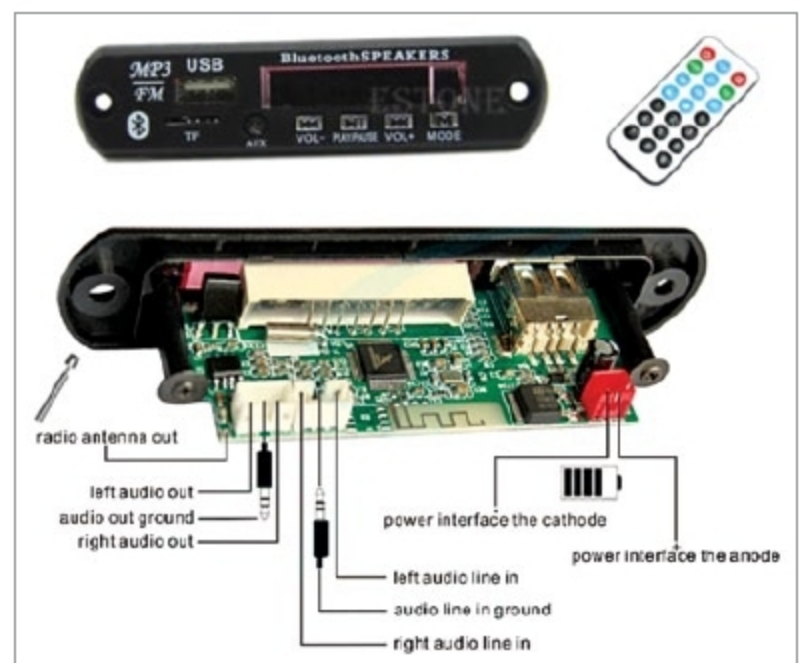


Fig. 3: VTF-108 Bluetooth module

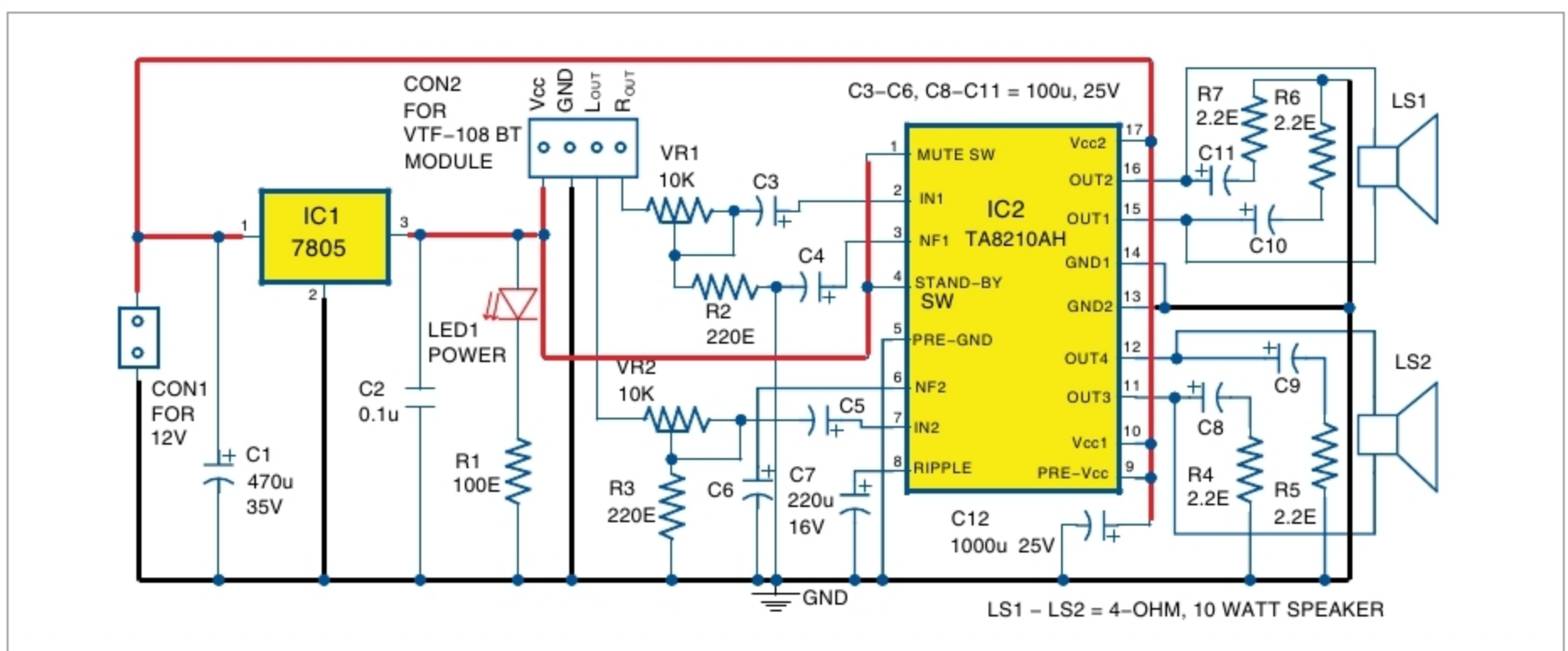


Fig. 2: Circuit diagram of 2-channel wireless Bluetooth audio amplifier